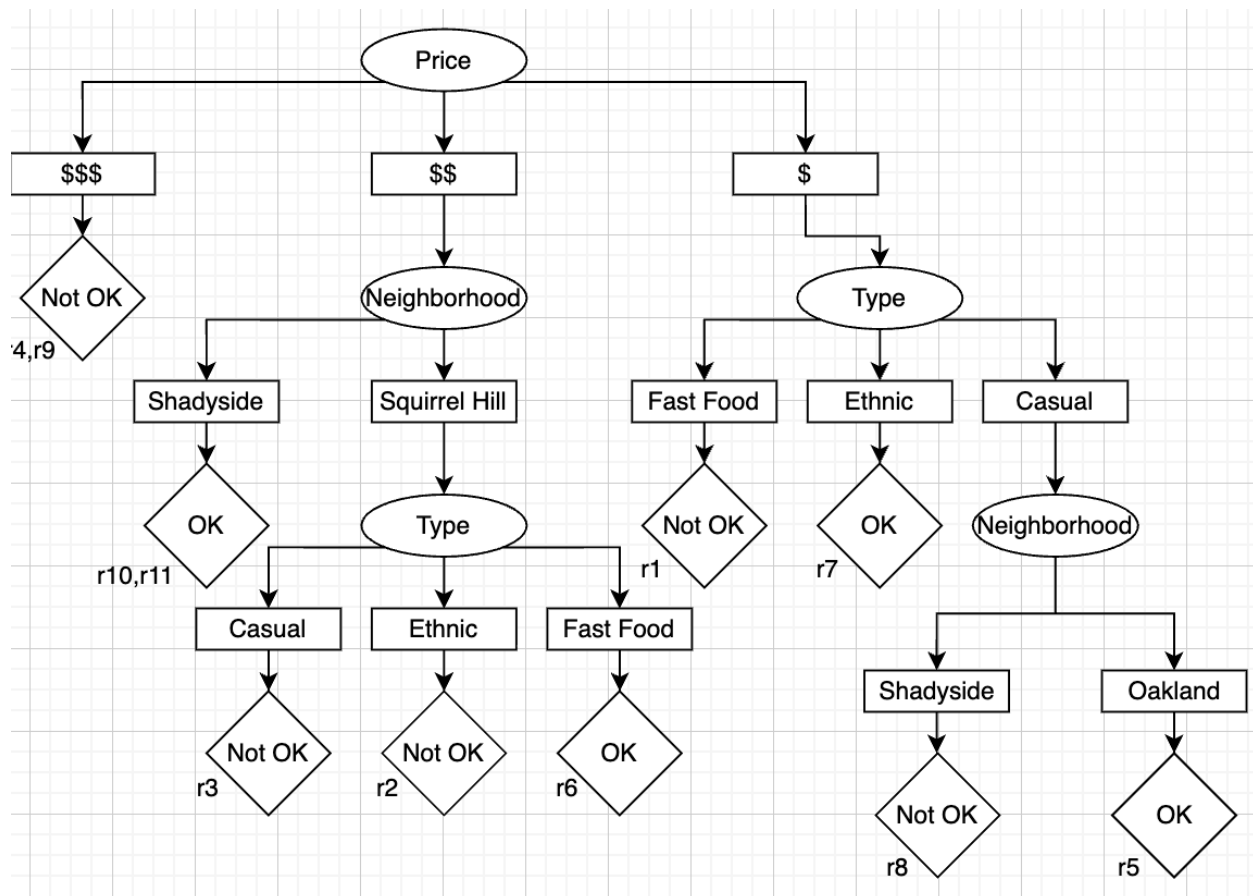


Homework 4

Problem 1.

(a) Using this data, build a decision tree to decide whether you would enjoy a particular restaurant or not, showing at each level how you decided which attribute to expand next based on the information gain of the entropy.



$\text{Info}(D) = - [(5/11)\log_2(5/11) + (6/11)\log_2(6/11)] = 0.994030211477$ entropy of dataset

Restriction:

$$\text{Vegetarian } H = -[(2/4)\log_2(2/4) + (2/4)\log_2(2/4)] = 1$$

$$\text{Gluten Free } H = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918295834054$$

$$\text{None } H = -[(2/4)\log_2(2/4) + (2/4)\log_2(2/4)] = 1$$

$$\text{all } H = (4/11)(1) + (3/11)(0.9182958) + (4/11)(1) = 0.9777170364$$

$$\text{Gain} = 0.994030211477 - 0.9777170364 = 0.0163131751$$

Neighborhood:

$$\text{Oakland } H = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918295834054$$

$$\text{Shadyside } H = -[(2/4)\log_2(2/4) + (2/4)\log_2(2/4)] = 1$$

Squirrel Hill $H = -[(2/4)\log_2(2/4) + (2/4)\log_2(2/4)] = 1$
all $H = (4/11)(1) + (3/11)(0.9182958) + (4/11)(1) = 0.9777170364$
Gain = $0.994030211477 - 0.9777170364 = 0.0163131751$

Price:

$\$ H = -[(2/4)\log_2(2/4) + (2/4)\log_2(2/4)] = 1$
 $\$ \$ H = -[(3/5)\log_2(3/5) + (2/5)\log_2(2/5)] = 0.970950594455$
 $\$ \$ \$ H = -[(2/2)\log_2(2/2)] = 0$
all $H = (4/11)(1) + (2/11)(0) + (5/11)(0.970950594455) = 0.8049775429$
Gain = $0.994030211477 - 0.8049775429 = 0.1890526686$

Type:

Fast Food $H = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918295834054$
Ethnic $H = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918295834054$
Casual Dining $H = -[(3/5)\log_2(3/5) + (2/5)\log_2(2/5)] = 0.970950594455$
all $H = (3/11)(0.918295834054) + (3/11)(0.918295834054) + (5/11)(0.970950594455)$
 $= 0.9422298161$
Gain = $0.994030211477 - 0.9422298161 = 0.0518003954$

Greatest gain was **Price**. Now we split on price:

$\$ \$ \$$ Branch:

end, because both r_4, r_9 are (not) OK=0

$\$ \$$ Branch:

We have $r_2, r_3, r_6, r_{10}, r_{11}$ left
info($\$ \$$) = $-[(3/5)\log_2(3/5) + (2/5)\log_2(2/5)] = 0.970950594455$
calculate gain of type, neighborhood, restriction:

Type:

Fast Food $H =$ only 1 so $H = 0$
Ethnic $H = 1$ (split even)
Casual Dining $H = 1$ (split even)
all $H = (2/5)(1) + (2/5)(1) + (1/5)(0) = 0.8$
Gain = $0.970950594455 - 0.8 = 0.1709505945$

Neighborhood

Shadyside $H =$ both 1 so $H = 0$
Squirrel Hill $H = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918295834054$
all $H = (3/5)(0.918295834054) + (2/5)(0) = 0.5509775004$
Gain = $0.970950594455 - 0.5509775004 = 0.4199730941$

Restriction

Vegetarian $H =$ only 1 so $H = 0$
Gluten Free $H =$ even split so $H = 1$
None $H =$ even split so $H = 1$
all $H = (2/5)(1) + (2/5)(1) + (1/5)(0) = 0.8$
Gain = $0.970950594455 - 0.8 = 0.1709505945$

Highest gain is Neighborhood so split on neighborhood:

Shadyside branch: done

squirrel hill branch:

$$\text{info}(\$ \$, \text{SH}) = -[(1/3)\log_2(1/3) + (2/3)\log_2(2/3)] = 0.918295834054$$

type:

$$\text{fast food } H = 0$$

$$\text{ethnic } H = 0$$

$$\text{casual } H = 0$$

$$\text{gain} = 0.918295834054 - 0 = 0.918295834054$$

restriction:

$$\text{GF } H = 0$$

$$\text{None } H = 1 \text{ even split}$$

$$\text{all } H = (2/3)(1) + (1/3)(0) = 0.6666666667$$

$$\text{gain} = 0.918295834054 - 0.6666666667$$

highest gain is type

branch on type

\$ branch:

We have r8, r7, r5, r1 left

$$\text{info}(\$) = -[(2/4)\log_2(2/4) + (2/4)\log_2(2/4)] = 1$$

type:

$$\text{Fast } H = 0$$

$$\text{casual } H = 1$$

$$\text{ethnic } H = 0$$

$$\text{all } H = (2/4)(1) = 0.5$$

$$\text{gain} = 1 - 0.5 = 0.5$$

neighborhood:

$$\text{oakland } H = 1$$

$$\text{SH } H = 0$$

$$\text{Shady } H = 0$$

$$\text{all } H = (2/4)(1) = 0.5$$

$$\text{gain} = 1 - 0.5 = 0.5$$

restriction:

$$\text{vegetarian } H = 1$$

$$\text{none } H = 0$$

$$\text{gf } H = 0$$

$$\text{all } H = (2/4)(1) = 0.5$$

$$\text{gain} = 1 - 0.5 = 0.5$$

all same gain, pick type for fun

fast food branch: done

ethnic branch: done

casual branch:

only two options, choose neighborhood:

oakland done

shadyside done

(b) You are now given data from five more restaurants: To which one would you go?

R12: No, I would not go because prev. tree says \$ Fast Food not OK

R13: Yes, because Shadyside \$\$ is good

R14: No, tree says \$ Fast Food bad

R15: Yes, because Shadyside \$\$ is good

R16: Yes, because \$ Ethnic is good

Based on my logic I would choose either R13, R15, or R16. However I would likely choose **R15** because Shadyside has multiple good restaurants, so I'm more biased to that.

(c) Out of curiosity, and to verify your decision tree accuracy, you decide to try them all. How good did your decision tree do? What is the test set error $E_{test}(h)$? What is the F1 score? To what do you attribute the results of your decision tree?

The only one I was wrong about was R16 out of the 5. $\frac{4}{5}=80\%$ of my predictions were correct!

$$TP = 2$$

$$FP = 1$$

$$TN = 2$$

$$FN = 0$$

$$\text{prec} = TP/(TP+FP) = 2/(2+1) = 2/3$$

$$\text{recall} = TP/(TP+FN) = 2/2$$

$$F1 = 2 * (\text{prec} * \text{recall}) / (\text{prec} + \text{recall}) = 2 * (2/3 * 2/2) / (2/3 + 2/2) = 0.8$$

$$\text{error}(h) = 0.8 = 80\%$$

$$F1 \text{ also} = 0.8$$

I attribute these results that there weren't enough rest. to have fully branching trees for everything, but overall it was pretty good.

Problem 2.

We want to predict if the person will play tennis in the three future days:

- **(8 points) Day 26: (Outlook=Sunny, Wind=Strong) $\rightarrow$ PlayTennis = ?**

$$P(\text{Yes}) = 12/25$$

$$P(\text{Outlook=Sunny} \mid \text{Yes}) = 4/12$$

$$P(\text{Wind=Strong} \mid \text{Yes}) = 4/12$$

$$\begin{aligned} P(\text{Yes}) * P(\text{Outlook=Sunny} \mid \text{Yes}) * P(\text{Wind=Strong} \mid \text{Yes}) &= 12/25 * 4/12 * 4/12 = \\ &= 0.0533333333 \end{aligned}$$

$$P(\text{No}) = 13/25$$

$$P(\text{Outlook=Sunny} \mid \text{No}) = 5/13$$

$$P(\text{Wind=Strong} \mid \text{No}) = 5/13$$

$$P(\text{No}) * P(\text{Outlook}=\text{Sunny} \mid \text{No}) * P(\text{Wind}=\text{Strong} \mid \text{No}) = 13/25 * 5/13 * 5/13 = 0.0769230769$$

Maximum is the No part ($0.0769230769 > 0.0533333333$)

PlayTennis = No

- **(8 points) Day 27: (Outlook=Overcast, Wind=Weak) → PlayTennis = ?**

$$P(\text{Yes}) = 12/25$$

$$P(\text{Outlook}=\text{Overcast} \mid \text{Yes}) = 4/12$$

$$P(\text{Wind}=\text{Weak} \mid \text{Yes}) = 8/12$$

$$P(\text{Yes}) * P(\text{Outlook}=\text{Overcast} \mid \text{Yes}) * P(\text{Wind}=\text{Weak} \mid \text{Yes}) = 12/25 * 4/12 * 8/12 = 0.1066666667$$

$$P(\text{No}) = 13/25$$

$$P(\text{Outlook}=\text{Overcast} \mid \text{No}) = 5/13$$

$$P(\text{Wind}=\text{Weak} \mid \text{No}) = 8/13$$

$$P(\text{No}) * P(\text{Outlook}=\text{Overcast} \mid \text{No}) * P(\text{Wind}=\text{Weak} \mid \text{No}) = 13/25 * 5/13 * 8/13 = 0.1230769231$$

Maximum is the No part ($0.1230769231 > 0.1066666667$)

PlayTennis = No

- **(8 points) Day 28: (Outlook=Rain, Wind=Weak) → PlayTennis = ?**

$$P(\text{Yes}) = 12/25$$

$$P(\text{Outlook}=\text{Rain} \mid \text{Yes}) = 4/12$$

$$P(\text{Wind}=\text{Weak} \mid \text{Yes}) = 8/12$$

$$P(\text{Yes}) * P(\text{Outlook}=\text{Rain} \mid \text{Yes}) * P(\text{Wind}=\text{Weak} \mid \text{Yes}) = 12/25 * 4/12 * 8/12 = 0.1066666667$$

$$P(\text{No}) = 13/25$$

$$P(\text{Outlook}=\text{Rain} \mid \text{No}) = 3/13$$

$$P(\text{Wind}=\text{Weak} \mid \text{No}) = 8/13$$

$$P(\text{No}) * P(\text{Outlook}=\text{Rain} \mid \text{No}) * P(\text{Wind}=\text{Weak} \mid \text{No}) = 13/25 * 3/13 * 8/13 = 0.0738461538$$

Maximum is the Yes part ($0.0738461538 < 0.1066666667$)

PlayTennis = Yes

Problem 3.

How is the point marked by “?” classified using Euclidean distance as the distance metric for $k = 1$, 2, and 3?

For $k=1$:

The "?" point is closest to either point 1 or 10 (equidistant?), although both are white. It would be classified as white.

For k=2:

The "?" point is closest to points 1 and 10 (equidistant?), and both are white. It would be classified as white.

For k=2:

The "?" point is closest to points 1 and 10 (equidistant?), and both are white, and also the black point 7. It would be classified as white because 2/3 of the nearest 3 points are white.

Are there any points in the training set that would be misclassified using k = 1? If so, identify them.

Using k=1, the following points are closest to another point that is in a different classification, which would cause misclassification of the point:

- Point 9 (should be black, would be white from point 14)
- Point 13 (should be white, would be black from point 18)
- Point 14 (should be white, would be black from point 9)
- Point 18 (should be black, would be white from point 13)

Come up with a simple distance metric that would properly classify all the points in the training set for k = 1.

To correctly classify each point with k=1, take only the distance across the x axis instead of the euclidean distance. This correctly classifies every point.

What happens when k = 5 using your distance metric?

Using k=5 with the x-axis only distance metric, we get correct classifications aside from the group of black points 7, 8, 9, which become white. This is because for any one point within the group of three, you need 5 others and only 2/5 of the surrounding points are black, so it defaults to white all of the time.

Problem 4.

Using MIN clustering:

I used Python to calculate the clustering (script attached at end) and got the following output:

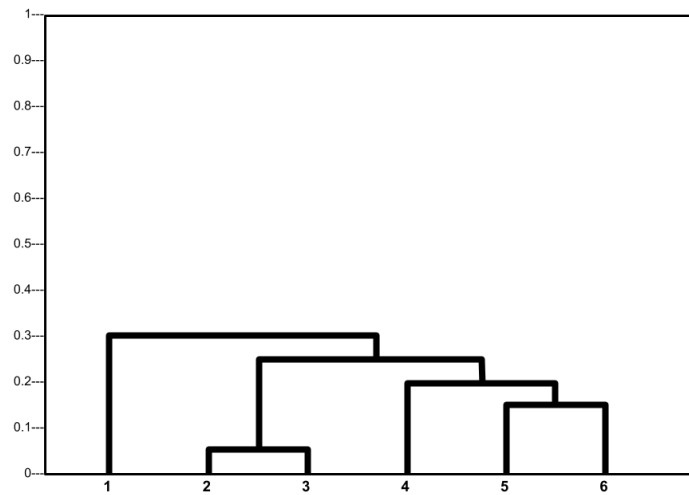
Merged 2 and 3 with min distance 0.05 to get [1, [2, 3], 4, 5, 6]

Merged 5 and 6 with min distance 0.15 to get [1, [2, 3], 4, [5, 6]]

Merged 4 and [5, 6] with min distance 0.20 to get [1, [2, 3], [4, [5, 6]]]

Merged [2, 3] and [4, [5, 6]] with min distance 0.25 to get [1, [[2, 3], [4, [5, 6]]]]

Merged 1 and [[2, 3], [4, [5, 6]]] with min distance 0.30 to get [[[1, [[2, 3], [4, [5, 6]]]]]]



Using MAX clustering:

I used the same Python to get:

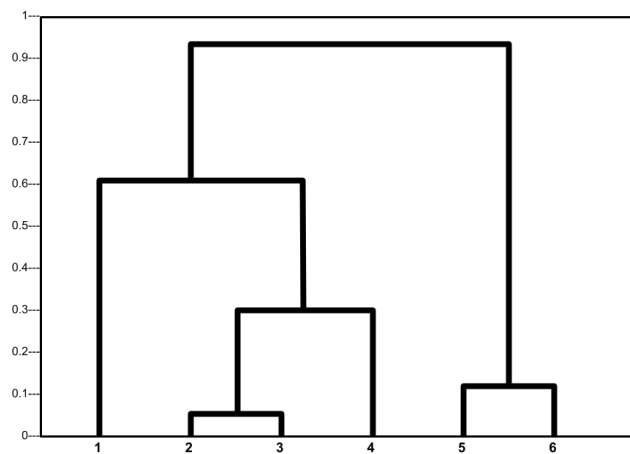
Merged 2 and 3 with distance 0.05 to get [1, [2, 3], 4, 5, 6]

Merged 5 and 6 with distance 0.15 to get [1, [2, 3], 4, [5, 6]]

Merged [2, 3] and 4 with distance 0.30 to get [1, [[2, 3], 4], [5, 6]]

Merged 1 and [[2, 3], 4] with distance 0.60 to get [[1, [[2, 3], 4]], [5, 6]]

Merged [1, [[2, 3], 4]] and [5, 6] with distance 0.95 to get [[[1, [[2, 3], 4]], [5, 6]]]



PYTHON SCRIPT:

```
import numpy as np

method = "MAX"

sim = np.array([
    [1.00, 0.70, 0.65, 0.40, 0.20, 0.05],
```

```

    [0.70, 1.00, 0.95, 0.70, 0.50, 0.35],
    [0.65, 0.95, 1.00, 0.75, 0.55, 0.40],
    [0.40, 0.70, 0.75, 1.00, 0.80, 0.65],
    [0.20, 0.50, 0.55, 0.80, 1.00, 0.85],
    [0.05, 0.35, 0.40, 0.65, 0.85, 1.00]
])

dists = 1 - sim

clusters = [1, 2, 3, 4, 5, 6]

def flatten(arr):
    if isinstance(arr, list):
        out = []
        for i in arr:
            out += flatten(i)
        return out
    else:
        return [arr]

def get_dist(ca, cb):
    all_a = flatten(ca)
    all_b = flatten(cb)

    extreme_dist = 1 if method == "MIN" else 0

    for a in all_a:
        for b in all_b:
            if method == "MIN":
                if dists[a-1, b-1] < extreme_dist:
                    extreme_dist = dists[a-1, b-1]
            else:
                if dists[a-1, b-1] > extreme_dist:
                    extreme_dist = dists[a-1, b-1]
    return extreme_dist

while len(clusters) > 1:
    candidates = []
    for i in range(len(clusters)):
        for j in range(i+1, len(clusters)):
            d = get_dist(clusters[i], clusters[j])
            candidates.append((d, clusters[i], clusters[j]))

```



```
ndist, tomerge_a, tomerge_b = min(candidates, key=lambda i: i[0])
new_cluster = [tomerge_a, tomerge_b]
clusters[clusters.index(tomerge_a)] = new_cluster
clusters.remove(tomerge_b)

print("Merged", tomerge_a, "and", tomerge_b, "with distance", str(round(ndist,
2)).ljust(4, "0"), "to get", clusters)
```